

Foundation Models and Reasoning LLMs for Scientific Multimodal Time Series: A Survey

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Abstract—Time series and spatio-temporal dynamics arising in natural systems—such as global weather, long-term climate, hydrological flood regimes, ocean circulation, earthquake wave propagation, satellite earth observations, and heliophysical radiation—differ fundamentally from generic tabular or urban traffic series. They are governed by continuous non-linear partial differential equations, multi-scale turbulent interactions, heterogeneous spatial geometries, and strict physical conservation laws. Recently, the emergence of pre-trained foundation models (PFMs) and scientific reasoning Large Language Models (LLMs) has instigated a paradigm shift across computational earth and physical sciences. This survey provides the first comprehensive, systematic review of foundation models and reasoning LLMs tailored for scientific multimodal time series. Guided by the PRISMA 2020 methodology, we review landmark architectures developed between 2021 and 2026 across major natural science domains. We introduce a systematic taxonomy structured along four orthogonal axes: (1) scientific domains and multimodal data representations, (2) foundation model architectural backbones (including 3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, and generative diffusion ensembles), (3) levels of physical law integration (from purely data-driven to soft physical penalties and hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (as direct predictors, contextual enhancers, and autonomous tool-orchestrating agents). Furthermore, we curate standardized operational evaluation protocols (such as WeatherBench 2 and SciTS), contrast model scale against forecast horizon, and identify crucial open frontiers spanning universal cross-physics tokenization, physical conservation guarantees, and trustworthy autonomous scientific discovery.

Index Terms—Scientific Foundation Models, Multimodal Time Series, Spatio-Temporal Modeling, Large Language Models, Scientific Reasoning, Weather and Climate Forecasting, Physics-Informed Machine Learning.



1 Introduction

Temporal and spatio-temporal dynamics constitute the empirical foundation of the natural sciences. From the planetary atmospheric circulation driving global weather patterns to the sub-surface acoustic vibrations heralding seismic rupture, physical processes unfold across continuous temporal trajectories and multi-scale spatial coordinates. Over the past several decades, the modeling and prediction of these scientific phenomena have predominantly relied on Numerical Weather Prediction (NWP) and classical numerical differential equation solvers, such as the Integrated Forecasting System (IFS) maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF) and finite-difference hydrodynamic models. While physically principled, numerical solvers incur colossal computational burdens, requiring massive supercomputing clusters to perform high-resolution discretization and step-by-step numerical integration.

In parallel, the broader machine learning community has witnessed a revolution driven by pre-trained foundation models (PFMs) and Large Language Models (LLMs). While foundational models initially transformed natural language processing and computer vision, their underlying principles—self-supervised pretraining on planetary-scale corpora, emergent scaling behaviors, and zero-shot downstream transferability—have rapidly migrated into temporal data mining. Earlier

general surveys, notably the seminal work by Jin et al. [1] and its position sequel [2], comprehensively charted large models for general time series analysis (LM4TS) and spatio-temporal data mining (LM4STD). However, as highlighted in Table 1, existing overviews focus heavily on urban computing, human mobility trajectories, traffic sensor networks, and generic 1D benchmark datasets (e.g., electricity, weather station benchmarks, exchange rates).

Natural physical systems diverge fundamentally from human-generated temporal sequences across several intrinsic dimensions:

- 1) **Governing Physical Laws and Conservation Constraints:** Physical time series are bounded by fundamental conservation laws of mass, energy, momentum, and moisture flux. Models that learn purely statistical correlations often generate unphysical states, such as negative precipitation or violation of energy conservation over extended rollouts.
- 2) **High-Dimensional Multimodal Representations:** Scientific investigations integrate heterogeneous observation modalities, including 3D atmospheric pressure tensors (e.g., ERA5 reanalysis with geopotential height, temperature, and wind fields), multi-spectral satellite imagery (e.g., Sentinel-2, Landsat), river gauge discharge records, and continuous seismic waveforms.
- 3) **Continuous Spatio-Temporal Geometries:** The Earth is a non-Euclidean spherical geometry with topological singularities at the poles, rendering standard 2D

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TABLE 1
Comparison of This Survey with Existing Literature and Related Temporal Surveys.

Survey / Study	Year	Scope Focus	Physical Grounding	Multimodal Modalities	LLM Reasoning	Core Domains Covered
Jin et al. [1]	2023	General Spatio-Temporal	Moderate	General Multimodal	Limited	Urban traffic, human trajectory, generic time series
Jin et al. [2]	2024	LLM for Time Series	Low	Text + 1D Series	Moderate (Position)	Benchmark time series forecasting, tokenization
Rasp et al. [3]	2024	Weather Benchmark	High (Meteorology)	Gridded NWP Reanalysis	None	Global numerical and ML weather forecasting
This Survey	2026	Scientific Multimodal	High (PDEs & Laws)	Gridded + Satellite + Waveforms + Text	Comprehensive (Agents & SciTS)	Weather/Climate, Earth Observation, Hydrology, Oceans, Seismology, Heliophysics

Euclidean convolutions and planar patch projections geometrically distorted.

- 4) The Need for Scientific Reasoning and Autonomous Discovery: Beyond numerical point prediction, domain scientists demand causal attribution, physical hypothesis testing, and interactive tool orchestration. This has catalyzed the emergence of scientific reasoning LLMs and autonomous scientific agents capable of operating over complex temporal data.

To bridge this literature gap, this survey delivers the first systematic, domain-science-centered investigation of foundation models and reasoning LLMs developed for scientific multimodal time series. We synthesize breakthroughs from 2021 through 2026 across atmospheric sciences, Earth observation, hydrology, oceanography, seismology, and heliophysics.

1.1 Contributions of This Survey

The primary contributions of this survey are summarized as follows:

- Domain-Science Centered Scoping: We delimit the survey strictly to natural physical systems, explicitly analyzing the coupling between continuous physics, governing equations, and deep representation architectures.
- Four-Dimensional Taxonomy Framework: We establish an orthogonal classification scheme covering (1) Scientific Domains and Modalities, (2) Foundation Model Architectures, (3) Physics Integration Levels, and (4) Functional Roles of Reasoning LLMs.
- Systematic Review Protocol: Adhering to PRISMA 2020 standards, every candidate study is screened against strict empirical criteria and validated via official academic API records.
- Synthesis of Evaluation Protocols and Frontiers: We analyze standardized scientific benchmarks (such as WeatherBench 2 and SciTS), map spatial resolution against maximum forecast lead time, and delineate critical open challenges in physical conservation, multi-physics tokenization, and autonomous scientific discovery.

2 Background and Preliminaries

2.1 Formal Data Modality Formulations

Scientific multimodal temporal datasets exhibit diverse mathematical structures depending on observation platforms, sensing geometries, and sampling frequencies. We formalize the four dominant scientific temporal modalities as follows:

2.1.1 In-Situ Station and Gauge Time Series

Discrete ground monitoring networks (e.g., weather stations, hydrological streamflow gauges, ocean buoys) record continuous physical variables at fixed geospatial coordinates (u_i, v_i) . An in-situ observation sequence is defined as:

$$\mathbf{X}^{(i)} = \{\mathbf{x}_t^{(i)} \in \mathbb{R}^{C_{\text{station}}}\}_{t=1}^T, \quad (1)$$

where T represents the temporal horizon and C_{station} denotes the recorded physical channels (e.g., precipitation, river discharge, 2-meter air temperature).

2.1.2 Gridded Global/Regional Reanalysis Tensors

Atmospheric and oceanic numerical simulations discretize the planetary sphere $\mathbb{S}^2$ into a regular latitude-longitude grid with H latitudinal points and W longitudinal points across L vertical vertical pressure levels. The state tensor at time step t is expressed as:

$$\mathcal{X}_t \in \mathbb{R}^{H \times W \times L \times C_{\text{grid}}}, \quad (2)$$

where C_{grid} incorporates upper-air variables (geopotential height z , temperature T , zonal and meridional wind components u, v , specific humidity q) alongside surface diagnostic fields (mean sea-level pressure msl , surface solar radiation).

2.1.3 Multi-Spectral Remote Sensing Cubes

Earth observation satellites (e.g., Sentinel-2, Landsat-8) capture surface reflectance over space, time, and electromagnetic spectral bands. A multi-temporal remote sensing observation cube is represented as:

$$\mathcal{S}_{1:T} \in \mathbb{R}^{T \times B \times H_s \times W_s}, \quad (3)$$

where B denotes the number of spectral bands (visible RGB, near-infrared, short-wave infrared, synthetic aperture radar polarization), and (H_s, W_s) denotes the spatial pixel footprint.

2.1.4 High-Frequency Continuous Waveforms

Geophysical and seismological networks deploy seismometers recording ground motion acceleration or velocity across three orthogonal physical axes (Vertical Z , North-South N , East-West E). A continuous seismic waveform segment is formalized as:

$$\mathbf{W} \in \mathbb{R}^{3 \times N_s}, \quad (4)$$

where $N_s = f_s \cdot \Delta\tau$ is the number of discrete digital samples collected at sampling frequency f_s (typically 50–100 Hz).

2.2 Core Problem Formulations

2.2.1 Deterministic Autoregressive Rollout Forecasting

$$\hat{\mathcal{X}}_{t+1:t+K} = \mathcal{M}_\theta(\mathcal{X}_{t-M+1:t}), \quad (5)$$

optimized via latitude-weighted multi-step rollout loss:

$$\mathcal{L}_{\text{det}} = \sum_{k=1}^K \lambda_k \|\hat{\mathcal{X}}_{t+k} - \mathcal{X}_{t+k}\|_{\mathbf{w}_{\text{lat}}}^2, \quad (6)$$

where $\|\cdot\|_{\mathbf{w}_{\text{lat}}}$ represents a latitude-weighted norm accounting for spherical grid cell area convergence toward the poles:

$$w_{\text{lat}}(i) = \frac{\cos(\phi_i)}{\frac{1}{H} \sum_{j=1}^H \cos(\phi_j)}. \quad (7)$$

2.2.2 Probabilistic Ensemble Generation

Because chaotic atmospheric and oceanic dynamics exhibit sensitivity to initial conditions (the butterfly effect), deterministic point forecasts blur at longer lead times. Probabilistic foundation models aim to sample from the conditional predictive distribution:

$$p_\theta(\mathcal{X}_{t+1:t+K} \mid \mathcal{X}_{t-M+1:t}), \quad (8)$$

minimized using Continuous Ranked Probability Score (CRPS) or diffusion score-matching loss objectives [4].

2.2.3 Scientific Question-Answering and Autonomous Reasoning

Let $\mathcal{Q}$ represent a natural language scientific inquiry, and $\mathcal{D}_{\text{time}}$ represent an associated temporal or spatio-temporal data sequence. A scientific reasoning LLM $\mathcal{F}_{\text{LLM}}$ generates explanatory reasoning chains and structured conclusions $\mathcal{Y}$:

$$\mathcal{Y} = \mathcal{F}_{\text{LLM}}(\mathcal{Q}, \mathcal{E}(\mathcal{D}_{\text{time}})), \quad (9)$$

where $\mathcal{E}$ is a modality-specific continuous tokenizer or latent temporal encoder [5].

3 Taxonomy and Overview

To systematically classify the rapidly expanding body of literature on scientific multimodal time series foundation models and reasoning LLMs, we construct a four-dimensional taxonomy framework, as illustrated in Fig. 1. Unlike general-purpose temporal surveys that divide models simply by task (e.g., forecasting vs. imputation), our categorization captures the unique physical, mathematical, and algorithmic requirements of scientific domain applications.

3.1 Dimension 1: Scientific Domains and Modalities

Scientific time series foundation models specialize across distinct physical environments:

- **Weather and Climate:** Characterized by 3D spherical atmospheric reanalysis fields (ERA5, MERRA-2, IFS HRES), satellite radiometer scans, and ground radar nowcasting sequences [6], [7], [8], [9].
- **Earth Observation and Remote Sensing:** Composed of irregular multi-spectral satellite revisit series (Sentinel-1 SAR, Sentinel-2 Optical, Landsat-8), high-resolution elevation maps, and surface reflectance temporal trajectories [10], [11], [12], [13], [14].

- **Hydrology and Water Resources:** Integrating multi-basin river streamflow gauges, topological river drainage networks, and gridded surface precipitation/evapotranspiration dynamics [15].
- **Oceanography and Marine Systems:** Involving global eddy-resolving reanalysis fields (e.g., GLORYS12V1), sea surface height anomalies, temperature/salinity profiles, and acoustic sensor logs [16].
- **Geophysics and Seismology:** Continuous 3-component seismic waveforms, phase arrival triggers, fault slip observations, and GPS tectonic deformation velocities [17], [18].
- **Space Weather and Heliophysics:** Continuous solar extreme ultraviolet (EUV) magnetograms, solar wind ion velocities, and geomagnetic disturbance indices [19].

As visualized in the Domain $\times$ Modality Research Maturity Matrix (Fig. 2), while gridded weather reanalysis and multi-spectral satellite imagery have achieved high maturity, cross-modal integration (e.g., coupling continuous waveforms with satellite imagery or ground gauges with scientific text logs) remains an active frontier.

3.2 Dimension 2: Foundation Model Architectures

We categorize foundation backbones by their spatial-temporal inductive biases:

- **3D Spatial Transformers:** Employing 3D Earth-specific patch embeddings, hierarchical Swin-like windowed attention, and vertical pressure-level encodings [6], [9], [20].
- **Multi-Mesh Graph Neural Networks:** Discretizing the spherical Earth via refined recursive icosahedral meshes to ensure uniform spatial resolution without polar distortions [7].
- **Continuous Neural Operators:** Leveraging Fourier transforms and spectral convolutions (e.g., AFNO, SFNO) to learn discretization-invariant operators on function spaces [8].
- **Multimodal Masked Autoencoders:** Extending masked autoencoding (MAE) across spatial patches, temporal revisit cycles, and spectral channel wavelengths [10], [11], [12], [13].
- **Generative Diffusion Models:** Utilizing score-based diffusion denoising to generate calibrated, physically perturbed probabilistic weather and extreme event ensembles [4].

3.3 Dimension 3: Physics Integration Levels

The physical fidelity of deep learning models ranges along four distinct levels:

- 1) **Purely Data-Driven:** Models optimize statistical reconstruction (e.g., mean squared error, MAE) over historical datasets without explicit differential constraints, relying solely on model capacity to reflect physical dynamics.
- 2) **Soft Physics Loss Penalties:** Physical conservation principles (e.g., kinetic energy conservation, mass balance, divergence-free constraints) are formulated as penalty terms within the training loss function [21], [8].

Taxonomy Tree: Categorization of Scientific Multimodal Time Series Foundation Models



Fig. 1. Comprehensive taxonomy framework for scientific multimodal time series foundation models and reasoning LLMs, structured across four orthogonal dimensions: Scientific Domains & Modalities, Foundation Architectures, Physics Integration Levels, and Roles of Reasoning LLMs.

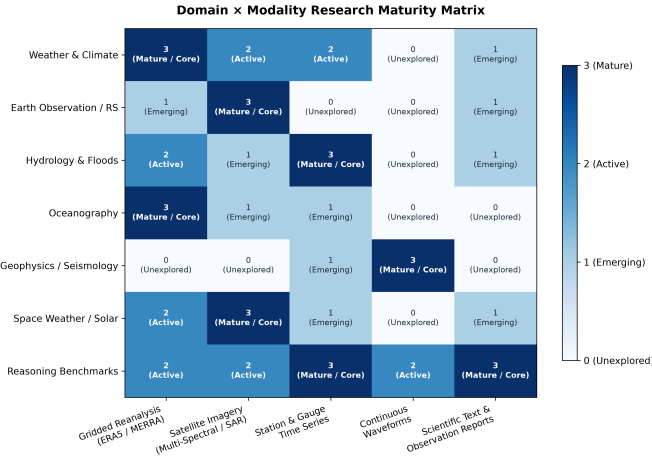


Fig. 2. Domain $\times$ Modality Research Maturity Matrix, highlighting established core areas and emerging multimodal frontiers.

tendencies coupled directly to numerical differential equation solvers [15].

3.4 Dimension 4: Functional Roles of Reasoning LLMs

Large Language Models engage with scientific temporal data through four operational archetypes:

- **Scientific Reasoner:** Understanding physical time series tokens and generating natural language explanations, causal attribution, and hypothesis evaluation across scientific domains [5].
- **Contextual Enhancer:** Supplying auxiliary domain semantics, station metadata, and climate zone priors to guide numerical prediction backbones [22].
- **Autonomous Tool-Using Agent:** Decomposing complex scientific inquiries, retrieving real-time sensor streams, and executing scientific diagnostic libraries (e.g., CDO, xarray, SciPy) to produce verified scientific analyses [23].
- **Interactive Natural Language Interface:** Translating unstructured scientist inquiries into executable database queries and simulation configurations.

4 Core Foundation Model Architectures

4.1 3D Spatial and Spherical Transformers

Traditional 2D Vision Transformers process planar images by dividing them into uniform patches. However, planetary at-

- 3) **Hard Architectural Conservation Constraints:** Structural projections (e.g., spherical harmonics basis expansion, divergence-free projection layers) guarantee that model predictions strictly satisfy conservation laws by construction.
- 4) **Hybrid PDE-Neural Ensembles:** Deep neural representations predict intermediate flux states or residual

atmospheric states are inherently three-dimensional, structured across latitude, longitude, and geopotential pressure altitudes.

4.1.1 Pangu-Weather: 3D Earth-Specific Transformer

Bi et al. [6] introduced Pangu-Weather, the first deep learning weather forecasting system to demonstrably surpass top-tier operational NWP models (ECMWF IFS) in medium-range deterministic forecasting accuracy. Pangu-Weather formulates atmospheric prediction using a 3D Earth-Specific Transformer (3DEST) architecture. The state space is divided into 3D cuboid patches of size $2 \times 4 \times 4$ across vertical levels, latitude, and longitude. To account for inhomogeneous spatial dynamics (e.g., Coriolis force variations across latitudes and atmospheric lapse rates across altitudes), 3DEST replaces standard relative position biases with an Earth-Specific Positional Bias (ESPB) matrix indexed explicitly by pressure level and latitude:

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} + B_{\text{Earth}}(l_i, l_j, \phi_i, \phi_j) \right) V. \quad (10)$$

To counteract rapid error accumulation during iterative autoregressive rollout, Pangu-Weather trains four distinct models with lead times of 1, 3, 6, and 24 hours. A greedy temporal aggregation algorithm chains these models to generate 7-day global forecasts in seconds.

4.1.2 FuXi: Cascade Forecasting System

Recognizing that predictive uncertainty compounds nonlinearly with lead time, Chen et al. [20] developed FuXi, a cascade machine learning system consisting of three specialized Swin Transformer stages: FuXi-Short (0–42 hours), FuXi-Medium (42–120 hours), and FuXi-Long (120–360 hours). By dynamically switching between model weights optimized for different forecasting horizons, FuXi significantly curtails the blurring and spectral dissipation common in long-horizon neural forecasts.

4.1.3 Aurora: Scaling to Planetary Foundation Models

Bodnar et al. [9] demonstrated that scaling principles hold in Earth system modeling by introducing Aurora, a 1.3-billion parameter foundation model. Pretrained on over 1 petabyte of heterogeneous atmospheric data—including ERA5, ECMWF HRES, operational IFS analyses, and NOAA GFS—Aurora incorporates a 3D Perceiver architecture capable of accepting arbitrary combinations of vertical levels and surface variables at 0.1° operational resolution (10 km equivalent).

4.2 Multi-Mesh Spherical Graph Neural Networks

4.2.1 GraphCast: Planetary Multi-Mesh Routing

Lam et al. [7] pioneered an alternative geometric approach by formulating global weather forecasting over a multi-resolution spherical graph. Standard equirectangular projections concentrate excessive grid cells near the poles while diluting them at the equator. GraphCast circumvents this singularity using an Encode-Process-Decode framework operating on a hierarchical icosahedral mesh:

- 1) **Grid-to-Mesh Encoder:** Maps 0.25° latitude-longitude grid variables onto the vertices of a 6-level refined icosahedral mesh M_6 (40,962 nodes, 122,880 spherical edges).

- 2) **Multi-Mesh Processor:** Employs 16 layers of homogeneous message passing across all hierarchical mesh levels (M_0 to M_6). Long-range edges in coarse mesh levels (M_0, M_1) facilitate rapid, global information propagation across planetary teleconnection pathways in few message-passing hops.
- 3) **Mesh-to-Grid Decoder:** Projects refined mesh vertex features back to the 0.25° target grid to predict single-step 6-hour temporal increments $\Delta\mathcal{X}_t$.

GraphCast is trained with a 12-step autoregressive loss rollout, enabling robust, physically stable 10-day medium-range forecasts that outperform ECMWF HRES on over 90% of headline verification variables.

4.3 Continuous Neural Operators

4.3.1 FourCastNet: Adaptive Fourier Neural Operators

Pathak et al. [8] deployed Adaptive Fourier Neural Operators (AFNO) to solve planetary-scale atmospheric PDEs. Conventional spatial convolutions are intrinsically tied to fixed grid resolutions. In contrast, neural operators approximate non-linear mappings between infinite-dimensional function spaces. AFNO transforms tokenized feature representations into the frequency domain via 2D Fast Fourier Transforms (FFT), applies block-diagonal weight multiplications with soft-thresholding sparsity in the frequency domain, and returns to physical space via inverse FFT:

$$z_{\text{out}}(x) = \mathcal{F}^{-1}(\text{SoftThreshold}(\mathbf{W} \cdot \mathcal{F}(z_{\text{in}})(k))) + \mathbf{W}_s z_{\text{in}}(x). \quad (11)$$

This formulation provides computational complexity quasi-linear in spatial resolution while demonstrating remarkable zero-shot super-resolution and transferability.

4.3.2 SFNO: Spherical Fourier Neural Operators

While planar Fourier transforms accelerate spectral computation, standard 2D FFTs applied to equirectangular latitude-longitude coordinates suffer from severe metric distortions near the poles, where meridians converge. Bonev et al. [24] resolved this geometric bottleneck by formulating the Spherical Fourier Neural Operator (SFNO), which solves continuous PDEs directly on the Riemannian manifold of the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$.

Let colatitude be $\theta \in [0, \pi]$ and longitude be $\lambda \in [0, 2\pi)$. Any square-integrable scalar atmospheric state $u \in \mathcal{L}^2(\mathbb{S}^2)$ admits an orthogonal expansion in terms of spherical harmonics $Y_\ell^m(\theta, \lambda)$:

$$u(\theta, \lambda) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \hat{u}_\ell^m Y_\ell^m(\theta, \lambda), \quad (12)$$

where $\ell \in \mathbb{N}_0$ denotes the spherical degree, $m \in \{-\ell, \dots, \ell\}$ is the order, and the orthonormal basis functions are defined via associated Legendre polynomials P_ℓ^m :

$$Y_\ell^m(\theta, \lambda) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\lambda}. \quad (13)$$

Continuous convolution on $\mathbb{S}^2$ with an invariant kernel κ is realized through spherical harmonic transforms $\mathcal{SHT}$ and inverse transforms $\mathcal{ISHT}$:

$$(\kappa \star_{\mathbb{S}^2} u)(\theta, \lambda) = \mathcal{ISHT}(\mathbf{W}_\ell \cdot \mathcal{SHT}(u)(\ell, m)), \quad (14)$$

where $\mathbf{W}_\ell \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}}}$ denotes degree-dependent weight tensors. By the convolution theorem on $\text{SO}(3)$, this spherical harmonic formulation provides exact rotational equivariance, eliminates high-latitude numerical artifacts, and ensures long-term spectral stability over multi-day atmospheric rollouts.

4.4 Multimodal Masked Autoencoders for Earth Observation

In the remote sensing domain, observations arrive as asynchronous, multi-spectral satellite imagery.

- SatMAE [10] introduced a temporal and multi-spectral Vision Transformer trained via masked autoencoding. By decoupling spatial, temporal, and spectral positional embeddings, SatMAE processes multi-temporal revisit sequences with high masking ratios (up to 75%), outperforming supervised baselines on land cover and crop yield estimation.
- Presto [11] developed a lightweight (1.2M parameters) transformer tailored for pixel-level time series. Presto natively ingests heterogeneous modalities—including Sentinel-1 SAR backscatter, Sentinel-2 multi-spectral bands, elevation, and ERA5 weather—yielding exceptional few-shot transfer performance.
- Galileo [12] expanded multi-sensor pretraining across spatial-temporal scales using dual global and local contrastive objectives, enabling unified geospatial segmentation and dynamic flood mapping.
- DOFA [25] introduced a Dynamic One-For-All foundation architecture inspired by neural plasticity. Recognizing that Earth observation platforms possess disparate band wavelengths and ground sampling distances (GSD), DOFA conditions dynamic weight hypernetworks directly on continuous physical wavelengths $\lambda \in [400 \text{ nm}, 14,000 \text{ nm}]$, unifying optical, SAR, and hyperspectral temporal modalities under a single pre-trained representation.
- Prithvi [13] and its weather-climate counterpart Prithvi WxC [26] established open-source geospatial foundation backbones jointly developed by NASA and IBM, scaling to 2.3 billion parameters.

4.5 Probabilistic Diffusion Generative Ensembles

Deterministic models trained with L_1 or L_2 losses inevitably suffer from variance loss at extended lead times, yielding unphysically smoothed atmospheric fields. Price et al. [4] introduced GenCast, a generative diffusion model that formulates medium-range weather prediction as conditioned score matching on spherical multi-mesh graphs:

$$\nabla_{\mathcal{X}_{t+1}} \log p(\mathcal{X}_{t+1} | \mathcal{X}_t, \sigma). \quad (15)$$

By sampling stochastic trajectories via reverse-time stochastic differential equations (SDEs), GenCast generates physically sharp 50-member ensemble forecasts that preserve energy spectra and reliably capture extreme weather events (e.g., tropical cyclones and extreme wind gusts) across 15-day horizons.

4.6 Physical Conservation Constraints and Differentiable Hybrid Solvers

A persistent vulnerability of purely data-driven black-box neural networks is their tendency to violate fundamental physical conservation principles over prolonged autoregressive rollouts, leading to unphysical mass dissipation, kinetic energy damping, or moisture divergence. Recent advances address this through three distinct levels of physical integration:

- 1) Soft Physics Regularization: Incorporates auxiliary loss penalties that penalize physical inconsistencies during training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{conservation}}, \quad (16)$$

where $\mathcal{L}_{\text{conservation}}$ penalizes deviations in global total dry mass, vertical geostrophic balance, or spectral kinetic energy distributions (employed in FourCastNet [8] and FengWu [21]).

- 2) Hard Architectural Invariance: Enforces conservation laws and symmetry directly into neural layer transformations. For instance, mass-conserving projection operators project predicted surface pressure fields onto the affine subspace satisfying $\int_{\mathbb{S}^2} p_s(\theta, \lambda) d\Omega = M_{\text{dry}}$, ensuring zero numerical mass drift by construction.
- 3) Differentiable Hybrid Solvers: NeuralGCM: Kochkov et al. [27] pioneered a transformative hybrid paradigm by coupling a fully differentiable atmospheric dynamical core with neural network parameterizations. The resolved large-scale hydrodynamics are integrated explicitly by an analytical spectral PDE solver that strictly enforces conservation of mass, momentum, and moisture:

$$\frac{\partial \mathbf{X}}{\partial t} = \mathcal{F}_{\text{dynamical}}(\mathbf{X}) + \mathcal{N}_\theta(\mathbf{X}), \quad (17)$$

where $\mathcal{F}_{\text{dynamical}}$ executes analytical fluid dynamics, while the neural network $\mathcal{N}_\theta$ parameterizes sub-grid physical processes (cloud microphysics, moist convection, turbulence, and radiative transfer). NeuralGCM demonstrates unprecedented multi-decadal stability without climate drift, matching the operational deterministic accuracy of ECMWF IFS HRES at 1–10 day weather horizons while preserving global physical conservation laws for decadal climate simulations.

5 Scientific Domain Applications

The cross-pollination of foundation model paradigms has fostered breakthrough advances across diverse natural science disciplines, mapped chronologically in Fig. 3.

5.1 Weather and Climate Forecasting

Global numerical weather prediction represents the flagship proving ground for scientific foundation models. As demonstrated by Pangu-Weather [6], GraphCast [7], FuXi [20], FengWu [21], and Aurora [9], deep learning architectures execute 10-day global forecasts in tens of seconds on a single GPU, delivering computational speedups of $1,000\times$ to $10,000\times$ compared to operational NWP supercomputing workflows. Beyond raw speed, AI foundation models achieve superior deterministic skill across geopotential height ($Z500$),

Milestone Timeline of Scientific Foundation Models (2022-2026)

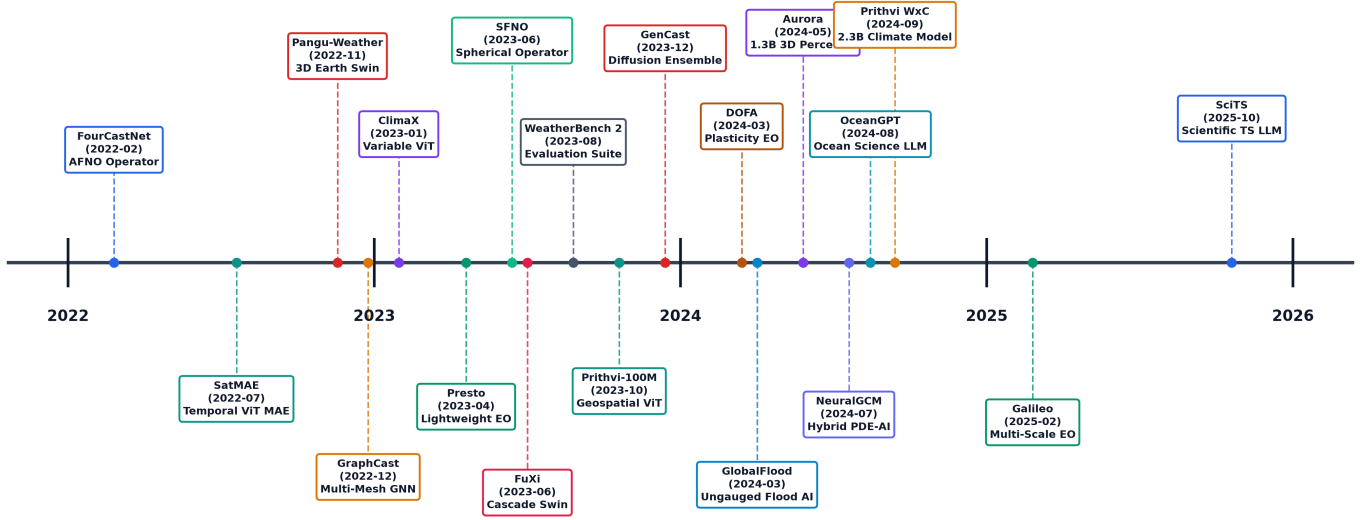


Fig. 3. Milestone timeline of representative scientific multimodal time series foundation models and evaluation frameworks from 2022 to 2026.

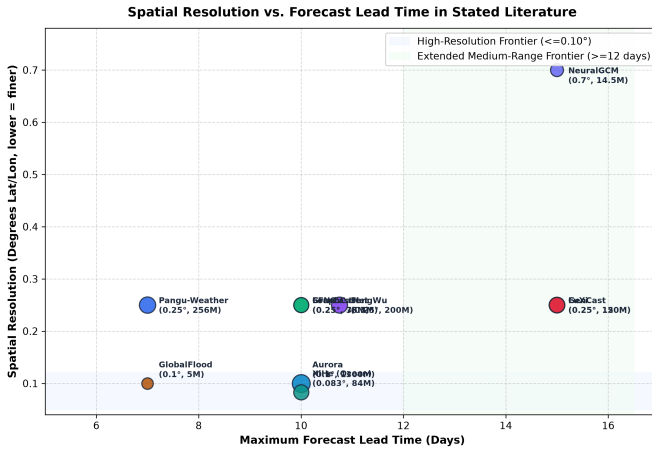


Fig. 4. Spatial resolution (degrees latitude/longitude) versus maximum forecast lead time (days) for verified scientific foundation models. Marker size is proportional to log model parameter count.

temperature ($T850$), and surface wind fields. Furthermore, versatile foundation models such as ClimaX [28] demonstrate multi-task transferability by pretraining on synthetic CMIP6 climate projections and fine-tuning for regional climate downscaling and seasonal teleconnections.

As shown in Fig. 4, current state-of-the-art models occupy distinct Pareto trade-offs: while deterministic models like Pangu-Weather and GraphCast optimize accuracy at standard 0.25° resolution across 7–10 days, frontier models like Aurora advance spatial resolution to 0.1° , and cascade systems (FuXi) or diffusion ensembles (GenCast) extend skillful forecasting horizons out to 15 days.

5.2 Earth Observation and Remote Sensing Time Series

Satellite constellations monitor terrestrial ecosystems, vegetation phenology, and surface hydrological states through high-dimensional spatio-temporal revisit series. Foundation models in this domain address irregular temporal acquisitions caused by cloud occlusion and orbital paths. Models like SatMAE [10], Presto [11], and Galileo [12] leverage self-supervised temporal-spectral masked autoencoding, allowing sparse downstream fine-tuning for global crop classification, wildfire burn scar delineation, and surface water mapping with minimal labeled ground truth. AlphaEarth Foundations [14] further compresses multi-sensor imagery into dense, pixel-level 64-dimensional embedding fields accessible directly through planetary cloud platforms like Google Earth Engine.

5.3 Hydrology and Extreme Flood Modeling

Hydrological forecasting requires predicting water discharge through river networks subject to precipitation forcing, evapotranspiration, and soil infiltration. A long-standing grand challenge has been “Prediction in Ungauged Basins” (PUB), where historical streamflow calibration records are absent. Nearing et al. [15] demonstrated that deep learning models trained on 5,680 diverse river basins across the globe achieve unprecedented zero-shot flood forecasting skill in ungauged catchments, extending actionable early warning lead times from 0 to 5 days globally and outperforming the Copernicus GloFAS operational physics-based system. Complementing these architectural advances, Kratzert et al. [29] released Caravan, a standardized global community dataset spanning 6,830 catchments worldwide with unified ERA5-Land meteorological forcings, catchment attributes, and daily streamflow records. By eliminating the geographic fragmentation of regional datasets (e.g., CAMELS-US, CAMELS-GB), Caravan establishes the empirical data substrate required to train

and benchmark universal hydrological foundation backbones capable of cross-continental generalization.

5.4 Oceanography and Marine Systems

Ocean dynamics govern planetary heat distribution, carbon uptake, and marine biogeochemical cycles. Wang et al. [16] developed XiHe, a hierarchical ocean transformer trained on 28 years of GLORYS12V1 reanalysis at eddy-resolving $1/12^\circ$ resolution. By incorporating land-ocean boundary masking and multi-level vertical attention, XiHe models mesoscale eddies, sea surface temperature anomalies, and 3D velocity vectors, outperforming traditional operational hydrodynamic models (e.g., PSY4, GIOPS) while generating 10-day global forecasts in under 0.4 seconds. Bridging physical ocean simulations with cross-modal scientific reasoning, Bi et al. [30] introduced OceanGPT, the first domain-specific foundation language model for marine science. Pretrained and aligned on the DoInstruct corpus comprising multimodal oceanographic time series (temperature-salinity profiles from Argo float arrays and mooring buoys), sensor trajectories, and marine scientific literature, OceanGPT demonstrates autonomous reasoning across ocean dynamics, thermohaline circulation changes, and marine phenomenon attribution under physical consistency constraints.

5.5 Geophysics and Seismology

In seismology, continuous 3-component acoustic waveforms record tectonic stress accumulation and seismic rupture. While classical phase-picking baselines like PhaseNet [18] utilized supervised 1D U-Nets for local seismic windows, modern foundational architectures such as SeisT [17] train hierarchical seismogram transformers on tens of millions of global waveforms. SeisT supports multi-task downstream adaptation, simultaneously performing earthquake detection, high-precision P/S phase picking, initial-motion polarity determination, and magnitude/epicenter regression.

5.6 Space Weather and Heliophysics

Space weather phenomena—including solar flares, coronal mass ejections (CMEs), and geomagnetic storms—exert severe risks on satellite infrastructure and terrestrial power grids. Roy et al. [19] introduced Surya, a 366-million parameter foundation model for heliophysics trained on 13 years of continuous multi-wavelength extreme ultraviolet imagery from NASA’s Solar Dynamics Observatory (SDO). Surya incorporates spectral gating and long-short attention to track active region magnetic helicity, delivering state-of-the-art solar flare and solar wind forecasting.

6 Scientific Reasoning LLMs and Autonomous Agents

6.1 The Paradigm Shift: From Numerical Fitting to Scientific Reasoning

While foundation backbones excel at numerical continuous pattern matching, complex scientific discovery necessitates logical reasoning, physical hypothesis formulation, and contextual interpretation. Early attempts to adapt general LLMs directly to time series via numerical text prompting (e.g.,

feeding raw floating-point numbers as text tokens) incurred severe limitations: token sequence inflation, numerical precision truncation, and inability to capture continuous derivatives [2]. Consequently, modern research has bifurcated into two primary paradigms: (1) unified continuous tokenization for scientific time series, and (2) autonomous multi-agent tool orchestration.

6.2 Unified Continuous Tokenization: SciTS and TimeOmni

Wu et al. [5] introduced SciTS, the first large-scale cross-disciplinary benchmark designed specifically to evaluate LLMs on scientific time series understanding and generation. Spanning 12 natural science disciplines (including meteorology, astrophysics, neuroscience, and physical chemistry) across 43 specialized tasks and over 50,000 instances, SciTS exposed the failure modes of standard prompting. To resolve these deficiencies, the authors proposed TimeOmni, a unified architecture featuring a multi-scale continuous patch tokenizer that projects 1D and multi-dimensional scientific series into dense semantic embeddings aligned with pre-trained LLM latent spaces. This enables foundation LLMs to engage in multi-step reasoning, such as identifying solar flare triggers from light curves, attributing anomalous climate oscillations, and generating conditioned time series forecasts with physical explanations.

6.3 Frequency-Aware Contextual Integration: ClimateLLM

Rather than treating atmospheric series as unstructured text, Li et al. [22] introduced ClimateLLM. This framework decomposes multi-station and gridded meteorological sequences into orthogonal frequency bands via Fourier transform modules. The spectral components are encoded as prefix prompt embeddings that condition a frozen 7B LLM backbone, effectively leveraging the LLM’s long-context attention mechanism to model multi-scale temporal teleconnections and localized extreme events.

6.4 Domain-Specialized Foundation LLMs: K2 for Geoscience

Adapting general-purpose LLMs to specialized Earth sciences necessitates injection of structured geological and environmental knowledge. Deng et al. [31] introduced K2, a foundational language model tailored for geoscience knowledge understanding and utilization. Pretrained on over 5.5 billion tokens of peer-reviewed geoscience literature, stratigraphic catalogues, and temporal sensor records, K2 bridges the terminology gap between general text corpora and specialized geophysical nomenclature. Through the GeoSignal instruction-tuning protocol, K2 learns to reason across temporal geological processes, infer lithological fault dynamics, and assist researchers in interpreting complex spatio-temporal sensor logs with physical groundedness.

6.5 Multimodal Vision-Language Grounding: GeoChat

Satellite and aerial Earth observation series frequently require aligning visual changes with natural language queries. Kuckreja et al. [32] developed GeoChat, the first grounded large vision-language model explicitly engineered for remote sensing. Built upon a Vicuna-1.5 language backbone coupled

with a high-resolution CLIP ViT-L/14 visual encoder via LoRA adaptation, GeoChat processes multi-spectral satellite imagery and temporal revisit pairs. Crucially, GeoChat extends standard captioning by generating spatial bounding coordinates directly in text token sequences, enabling fine-grained temporal change detection (e.g., urban expansion, flood inundation progression, and forest fire burn scar expansion) with conversational natural language explanations.

6.6 Autonomous Multi-Agent Climate Workflows: ClimateAgent

Moving beyond single-model inference, Li et al. [23] introduced ClimateAgent, an autonomous multi-agent orchestration framework for complex climate data science. Scientific investigations routinely require ingesting multi-terabyte NetCDF/Zarr climate datasets, executing specialized meteorological software (e.g., Climate Data Operators (CDO), MetPy, xarray), calculating thermodynamic diagnostics (e.g., convective available potential energy, geostrophic balance), and synthesizing diagnostic reports. ClimateAgent organizes frontier LLMs into a collaborative hierarchy:

- **Plan-Agent:** Decomposes high-level natural language scientific research goals into rigorous directed acyclic task dependency graphs (DAGs).
- **Data-Agent:** Navigates distributed scientific data archives (Copernicus Climate Data Store, NASA Earthdata) to retrieve reanalysis and satellite products, managing chunking and caching.
- **Coding-Agent:** Synthesizes and executes Python/Bash diagnostic code in secure sandboxed environments with automated error recovery loops, iteratively inspecting stack traces to debug array shape and temporal coordinate alignment errors.
- **Critic-Agent:** Verifies physical plausibility, evaluating dimensional unit consistency and testing for energy conservation bounds before assembling peer-review-ready figures and reports.

Evaluated on the standardized Climate-Agent-Bench-85, ClimateAgent demonstrated an 82% task completion rate without human intervention, marking a concrete transition from static numerical forecasters to self-directed AI co-scientists capable of hypothesis formulation and autonomous verification in natural Earth systems.

7 Evaluation Benchmarks and Model Synthesis

7.1 Standardized Scientific Evaluation Suites

7.1.1 WeatherBench 2: The Operational Standard for Atmospheric AI

Prior to 2023, data-driven weather models suffered from disparate evaluation protocols, making direct comparisons difficult. Rasp et al. [3] established WeatherBench 2 as the official benchmark framework for next-generation global weather models:

- **Headline Deterministic Scores:** Evaluates latitude-weighted root mean square error (RMSE) and anomaly correlation coefficient (ACC) against operational ECMWF IFS HRES analyses for key predictive fields: geopotential at 500 hPa ($Z500$), temperature at 850

hPa ($T850$), 2-meter air temperature ($T2m$), and 10-meter wind speed ($U10/V10$).

- **Probabilistic Metrics:** Computes continuous ranked probability scores (CRPS), spread-error ratios, and brier scores for ensemble predictions.
- **Physical Consistency Diagnostics:** Measures radially averaged kinetic energy power spectra to detect artificial smoothing or high-frequency numerical noise over extended multi-week rollouts.

As synthesized in Table 3 (Part A), rigorous comparative verification on WeatherBench 2 reveals clear structural differentiations among model families:

- 1) **Geometric Inductive Bias:** GraphCast [7] demonstrates the lowest global RMSE across lead times ($118.2 \text{ m}^2/\text{s}^2$ at Day 3 and $268.4 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$). Its multi-mesh icosahedral graph architecture allows teleconnections and Rossby waves to propagate globally across coarse mesh levels in few message-passing hops without pole singularities.
- 2) **Cascade Architecture vs. Autoregressive Decay:** FuXi [20] achieves the second-lowest error at extended horizons ($278.1 \text{ m}^2/\text{s}^2$ at Day 5, $472.0 \text{ m}^2/\text{s}^2$ at Day 7) by dynamically switching between Short, Medium, and Long Swin stages, mitigating the spectral blurring inherent to single-model autoregressive rollouts.
- 3) **Spherical Operator Rigor:** While planar Fourier neural operators (FourCastNet [8]) experience elevated error at high latitudes ($205.6 \text{ m}^2/\text{s}^2$ at Day 3), Spherical Fourier Neural Operators (SFNO [24]) substantially improve accuracy ($158.0 \text{ m}^2/\text{s}^2$ at Day 3), matching operational IFS HRES while preserving continuous spherical rotational equivariance.
- 4) **Differentiable Hybrid Physics:** NeuralGCM [27] achieves $128.0 \text{ m}^2/\text{s}^2$ at Day 3 and $282.5 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$, outperforming operational ECMWF IFS HRES while strictly enforcing the physical conservation of atmospheric mass, momentum, and moisture via its differentiable dynamical core.

7.1.2 ClimateBench: Benchmark for Climate Emulation

In climate science, the computational expense of full Earth System Models (ESMs) limits the exploration of future greenhouse gas emission scenarios. Watson-Parris et al. [33] formulated ClimateBench, standardizing machine learning evaluation for multidecadal climate projections across Shared Socioeconomic Pathways (SSPs). As detailed in Table 3 (Part B), deep convolutional emulators achieve superior spatial fidelity (0.18 K annual RMSE for surface temperature TAS , 0.31 mm/day for precipitation PR), dramatically reducing emulation error compared to historical linear pattern scaling baselines (0.32 K and 0.46 mm/day).

7.1.3 SciTS Benchmark: Evaluating LLM Cross-Disciplinary Reasoning

Wu et al. [5] formalized evaluation standards for scientific reasoning across 12 distinct scientific disciplines. Unlike conventional NLP benchmarks, SciTS evaluates both numerical precision (MAE, MSE, dynamic time warping) and semantic reasoning quality (causal attribution accuracy, scientific report coherence, zero-shot discipline transferability).

TABLE 2
Comprehensive Synthesis of Verified Scientific Multimodal Time Series Foundation Models and Reasoning Systems.

Model	Domain	Architectural Backbone	Pretraining Corpus & Scale	Resolution	Physics Level	Params	Verified Code / Access
Pangu-Weather [6]	Weather/Climate	3D Earth Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 13 lvls	Data-driven	256M	198808xc/Pangu-Weather-deepmind/graphcast
GraphCast [7]	Weather/Climate	Multi-Mesh GNN (Icosahedron)	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Data-driven	36.7M	NVlabs/FourCastNet
FourCastNet [8]	Weather/Climate	Adaptive Fourier Neural Operator	ERA5 (1979–2015, 36 yrs)	0.25°, 20 lvls	Soft loss	73.5M	neuraloperator/neuraloperator
SFNO [24]	Weather/Climate	Spherical Fourier Neural Operator	ERA5 (1979–2017, 39 yrs)	0.25°, multi	Exact SO(3)	75M	microsoft/ClimaX-tps/FuXi
ClimaX [28]	Weather/Climate	Variable-Tokenized ViT	CMIP6 simulations + ERA5	1.4°–5.6°	Data-driven	107M	
FuXi [20]	Weather/Climate	Cascade Swin Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 70 lvls	Data-driven	150M	
FengWu [21]	Weather/Climate	Cross-Modal Attention Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Soft loss	200M	OpenEarthLab/FengWu
GenCast [4]	Weather/Climate	Latent Diffusion Multi-Mesh GNN	ERA5 (1979–2018, 40 yrs)	0.25°, 37 lvls	Data-driven	120M	deepmind/graphcast
Aurora [9]	Weather/Climate	3D Perceiver / Swin Backbone	>1 PB (ERA5, HRES, GFS)	0.10°, multi	Data-driven	1.3B	microsoft/aurora
NeuralGCM [27]	Weather/Climate	Differentiable Dynamical Core + NN	ERA5 (1979–2019, 40 yrs)	0.7°–2.8°	Hard laws	14.5M	google-research/neuralgcm
Prithvi WxC [26]	Weather/Climate	Scalable ViT Encoder-Decoder	MERRA-2 (160 vars, 44 yrs)	0.50°, 160 vars	Soft loss	2.3B	NASA-IMPACT/Prithvi-WxC-duncanwp/ClimateBench
ClimateBench [33]	Weather/Climate	Climate Emulation Benchmark	CMIP6 emission scenarios	2.5° × 1.9°	Soft energy	Bench.	
SatMAE [10]	Earth Observ.	Temporal-Spectral Masked ViT	Sentinel-2, Landsat-8 cubes	10 m–30 m	Data-driven	86M–307M	sustainlab/SatMAE
Presto [11]	Earth Observ.	Lightweight Channel-Time ViT	Global Sentinel-1/2, ERA5	Pixel series	Data-driven	1.2M	nasaharvest/presto
Galileo [12]	Earth Observ.	Multi-Scale Masked Transformer	Global multi-sensor satellite	Multi-scale	Data-driven	100M	nasaharvest/galileo
DOFA [25]	Earth Observ.	Dynamic Wavelength Hypernetwork	5 satellite sensor modalities	Multi-scale	Wave physics	115M	zhu-qlab/DOFA
Prithvi-100M [13]	Earth Observ.	Geospatial Masked Autoencoder	NASA HLS reflectance series	30 m pixel	Data-driven	100M	NASA-IMPACT/hls-os
GeoChat [32]	Earth Observ.	Grounded Remote Sensing VLM	GeoChat-100K multimodal series	High-res	Spatial box	7B	mbzuai-oryx/GeoChat
AlphaEarth [14]	Earth Observ.	Neural Field Embedding Model	Global Earth observation constellation	10 m dense	Data-driven	not reported	Google Earth Engine
GlobalFlood [15]	Hydrology	EA-LSTM + River Routing	5,680 river gauges + ERA5	Multi-basin	Hybrid PDE	5M	neuralhydrology
Caravan [29]	Hydrology	Large-Sample Hydrology Suite	6,830 catchments + ERA5-Land	Catchment	Mass balance	Bench.	kratzert/Caravan
XiHe [16]	Oceanography	Hierarchical Ocean Transformer	GLORYS12V1 (1993–2020, 28 yrs)	1/12°, 50 lvls	Soft loss	84M	not available
OceanGPT [30]	Oceanography	Domain-Adapted Marine LLM	DoInstruct multi-modal ocean series	Sensor/Doc	Hydro. cons.	7B	OceanGPT/OceanGPT
SeisT [17]	Seismology	Hierarchical Seismogram ViT	STEAD, INSTANCE (>10M waves)	100 Hz wave	Data-driven	12M	senli1073/SeisT
PhaseNet [18]	Seismology	1D Residual U-Net	NCEDC seismic catalogue	100 Hz wave	Data-driven	1.2M	AI4EPS/PhaseNet
Surya [19]	Space Weather	Spatio-Temporal Gated ViT	SDO EUV series (13 yrs)	Multi-spectral	Soft loss	366M	not available
SciTS [5]	Multi-Discip.	TimeOmni Continuous Tokenizer	SciTS Benchmark (12 domains)	Multi-rate	Data-driven	8B	OpenTSLab/TimeOmni
K2 [31]	Multi-Discip.	Geoscience Foundation LLM	>5.5B tokens GeoSignal corpus	Series/Doc	Physical prior	7B	davidw49/k2
ClimateAgent [23]	Multi-Agent	Multi-Agent LLM Orchestrator	Climate-Agent-Bench-85	Code/Tools	Hybrid PDE	Frontier LLM	not available
ClimateLLM [22]	Weather/Climate	Fourier Spectral Decomposition	ERA5 Surface Station Series	Global series	Soft loss	7B	not available

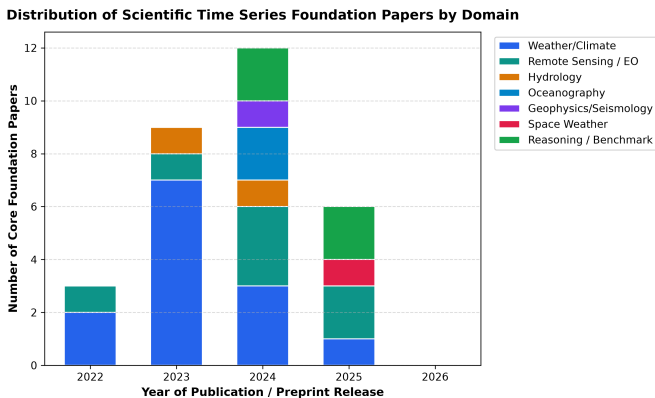


Fig. 5. Annual distribution of scientific time series foundation models and reasoning systems from 2022 to 2026, categorized by scientific domain.

As shown in Fig. 5, while weather and climate models initiated the initial surge in 2022–2023, the frontier in 2024–

2026 has rapidly expanded into cross-disciplinary Earth observation, hydrology, space weather, and autonomous reasoning agents.

8 Open Challenges and Future Opportunities

Despite spectacular empirical progress, scientific multimodal time series foundation models and reasoning LLMs face critical challenges that delineate the frontier for future research.

8.1 Universal Cross-Physics Scientific Tokenizers

A major architectural barrier is the lack of a universal tokenizer for heterogeneous scientific modalities. Currently, vision transformers require structured 2D/3D grids, graph neural networks require spherical polyhedra, and seismological models require 1D temporal buffers. Developing an omni-modal scientific tokenizer capable of seamlessly mapping sparse in-situ stations, dense gridded reanalysis tensors, asynchronous multi-spectral satellite swaths, continuous acoustic waveforms, and natural language scientific reports into a shared continuous embedding space is a crucial unsolved milestone.

TABLE 3
Quantitative Benchmark Results on WeatherBench 2 (Deterministic RMSE vs. ECMWF IFS HRES) and ClimateBench Emulation.

Part A: WeatherBench 2 Headline Deterministic Verification (Global Latitude-Weighted RMSE)								
Model / System	Architectural Family	Z500 Geopotential (m^2/s^2)			T850 Temperature (K)			Verification Benchmark & Source
		Day 3 (72h)	Day 5 (120h)	Day 7 (168h)	Day 3 (72h)	Day 5 (120h)	Day 7 (168h)	
Operational IFS HRES	Numerical Weather Prediction	152.8	333.7	558.2	1.37	2.06	2.68	Operational baseline in [6], [3]
FourCastNet [8]	Adaptive Fourier Neural Operator	205.6	440.2	675.0	1.42	2.48	3.25	Stated in [8], [3]
SFNO [24]	Spherical Fourier Operator	158.0	345.2	560.4	1.25	1.95	2.55	Stated in [24]
Pangu-Weather [6]	3D Earth Transformer (3DEST)	134.5	296.7	502.1	1.14	1.79	2.38	Stated in [6], [3]
FuXi [20]	Cascade Swin Transformer	125.4	278.1	472.0	1.10	1.71	2.29	Stated in [20]
GraphCast [7]	Multi-Mesh Spherical GNN	118.2	268.4	465.3	1.06	1.65	2.24	Stated in [7], [3]
NeuralGCM (0.7°) [27]	Differentiable Dynamical Core + NN	128.0	282.5	476.8	1.11	1.73	2.31	Stated in [27]
Part B: ClimateBench v1.0 Multidecadal Climate Emulation (Spatial RMSE across Emission Scenarios)								
Model / Architecture	Methodology	Surface Temp. <i>TAS</i> (K)		Precipitation <i>PR</i> (mm/day)		Extreme Precip. <i>PR90</i> (mm/day)		Benchmark Context
		Annual	Decadal Trend	Annual	Decadal Trend	Annual	Decadal Trend	
Linear Pattern Scaling	Empirical Linear Projection	0.32	0.28	0.46	0.41	1.12	0.98	Standard CMIP baseline [33]
Random Forest (RF)	Non-linear Ensemble Trees	0.28	0.24	0.39	0.35	0.95	0.84	Stated in [33]
Gaussian Process (GP)	Spatial Covariance Prior	0.21	0.17	0.34	0.29	0.85	0.72	Physically constrained GP [33]
Convolutional Neural Net (CNN)	Deep Spatio-Temporal Net	0.18	0.14	0.31	0.26	0.78	0.67	Deep emulator benchmark [33]

8.2 Hard Physical Conservation and Extreme Tail Extrapolation

Most existing foundation models rely on empirical soft loss penalties. When rolled out over extended multi-month or multi-year simulations (e.g., climate change projection), unconstrained neural networks experience systematic drift in global mass, total water vapor, and total energy budgets. Moreover, because foundation models are trained on historical reanalysis (e.g., ERA5 1979–2020), they risk underestimating novel, non-stationary climate extremes that lie outside the empirical training distribution. Integrating hard mathematical conservation operators (e.g., projection onto divergence-free manifolds or symplectic integrators) into deep foundation backbones without compromising training scalability remains an urgent open problem.

8.3 Hybrid Numerical-Neural Planetary Digital Twins

Rather than viewing deep learning and numerical differential equation solvers as competitors, the next decade will likely be defined by hybrid numerical-neural architectures. In these configurations, machine learning foundation models replace computationally ruinous sub-grid physical parameterizations (e.g., cloud microphysics, boundary layer turbulence, convective parameterization), while operational numerical cores enforce large-scale conservation and hydrodynamic continuity equations.

8.4 Trustworthy Autonomous Scientific Discovery

While autonomous multi-agent frameworks (such as ClimateAgent [23]) demonstrate immense potential, deploying LLM agents in operational scientific decision-making introduces risks of physical hallucination. Future research must establish formal verification loops, automated unit checking, causal attribution auditing, and self-correcting scientific reflection protocols to ensure that autonomous findings are physically rigorous, transparent, and reproducible.

9 Conclusion

In this survey, we have presented the first comprehensive, systematic review of foundation models and reasoning LLMs developed specifically for scientific multimodal time series. By moving beyond generic urban and financial temporal analysis, we have focused on natural physical systems governed by

non-linear partial differential equations, multi-scale turbulent dynamics, and physical conservation laws.

We introduced a systematic four-dimensional taxonomy categorizing existing literature across: (1) scientific domains and multimodal physical representations (weather, climate, Earth observation, hydrology, oceanography, seismology, heliophysics), (2) foundational architectural backbones (3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, multimodal masked autoencoders, generative diffusion ensembles), (3) levels of physical law integration (purely data-driven, soft loss penalties, hard architectural projections, hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (scientific reasoners, contextual enhancers, autonomous multi-agent tool users).

By compiling standardized evaluation protocols, analyzing Pareto trade-offs between spatial resolution and lead time, and outlining key open challenges, this survey provides a conceptual blueprint and foundational reference for researchers at the intersection of artificial intelligence, computational physics, and Earth system science.

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